GIRISH VUYYURU

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Senior AI Engineer | Agentic AI · Generative AI · LLMs | LangGraph · LangChain · MCP · RAG · RAGAS Evaluation | Python · FastAPI · AWS · Azure · GCP | Docker · Kubernetes · CI/CD

PROFESSIONAL SUMMARY

Senior AI Engineer with 6+ years building and deploying production AI solutions, with daily hands-on Claude Code and AI-assisted engineering and agentic AI: multi-agent systems: agent orchestration with LangGraph and LangChain (planning, tool calling, shared state, memory, human-in-the-loop), MCP servers, agent-to-system integrations and user-facing agent interactions, with RAG, LLMs and Python/FastAPI services. Delivered production-grade Generative AI and Agentic AI systems across financial services, federal, and enterprise environments. Deep expertise in LLMs, multi-agent orchestration, LangChain, LangGraph, MCP/FastMCP, RAG, GraphRAG, Prompt Engineering, and Context Engineering — designing agents with tool calling, conditional routing, shared state, memory, and human-in-the-loop workflows. Production track record across OpenAI, Azure OpenAI, and AWS Bedrock LLM integrations with vector databases (pgvector, Pinecone, ChromaDB, FAISS). Full-stack Python engineer: FastAPI/Flask REST APIs, microservices, Docker/Kubernetes containerization. AI Model Evaluation and Monitoring practitioner using RAGAS, MLflow, Prometheus, and Grafana. AWS and Azure cloud-native AI platform deployments with full CI/CD automation.

TECHNICAL SKILLS

Generative AI & LLMs: OpenAI APIs, Azure OpenAI, AWS Bedrock, LLMs, Large Language Models, Prompt Engineering, Context Engineering, Few-Shot Learning, Fine-Tuning, LLM Evaluation, RAGAS

Agentic AI & Orchestration: LangChain, LangGraph, LlamaIndex, FastMCP/MCP, AI Agents, Multi-Agent Systems, Agent Orchestration, Tool Calling, Function Calling, Conditional Routing, Shared State, Memory, Human-in-the-Loop

RAG & Vector Search: RAG, GraphRAG, Retrieval-Augmented Generation, Embeddings, Semantic Search, pgvector, Pinecone, ChromaDB, FAISS, Weaviate, Metadata Filtering, Reranking

Python & APIs: Python, FastAPI, Flask, Django, REST APIs, Microservices, Async Processing, OpenAPI/Swagger, Pydantic, Kafka, Streaming

AI/ML Integration: Scikit-learn, PyTorch, TensorFlow, XGBoost, Hugging Face Transformers, NLP, MLflow, Model Evaluation, Model Monitoring, Drift Detection, Production Deployment

Cloud Platforms: AWS (Lambda, SageMaker, ECS, EKS, Bedrock, API Gateway, S3, RDS, DynamoDB, IAM, CloudWatch, SQS, SNS), Azure (AML, Databricks, AI Search, ADF, ADLS Gen2, AKS)

DevOps & Observability: Docker, Kubernetes, GitHub Actions, Azure DevOps, Jenkins, CI/CD, Terraform, OpenTelemetry, Prometheus, Grafana, Structured Logging

Databases: PostgreSQL, MySQL, Redis, DynamoDB, MongoDB, Snowflake, Snowflake Cortex, Microsoft Fabric, pgvector, Pinecone, ChromaDB

Additional Languages & Data: Java, Golang, Angular, GraphQL, SQL Server, DB2, R

PROFESSIONAL EXPERIENCE

Agentic AI Developer | Credit Union of Texas | Dec 2025 – Present | Houston, TX

- **Agentic AI & multi-agent orchestration:** Designed and deployed production multi-agent LangGraph systems with tool calling, conditional routing, shared state, memory, and human-in-the-loop workflows — reduced AI feature development cycles by enabling teams to compose agents from shared reusable components for financial services applications.
- **MCP/FastMCP server engineering:** Architected and deployed FastMCP/MCP servers as enterprise AI infrastructure — standardized tool exposure and function calling across 5+ applications, enabling reusable LLM integrations and self-service AI capabilities for developer and business teams.
- **RAG & GraphRAG pipeline development:** Built end-to-end RAG and GraphRAG pipelines — document ingestion, embeddings generation, semantic search with pgvector/Pinecone/ChromaDB, metadata filtering, reranking, and grounded LLM response generation via OpenAI and AWS Bedrock for financial document intelligence.
- **Prompt Engineering & Context Engineering:** Designed advanced prompt engineering and context engineering strategies — few-shot templates, chain-of-thought, structured output constraints, and dynamic context injection — achieving measurable improvements in LLM response accuracy and grounding across production GenAI applications.
- **LLM model evaluation & monitoring:** Implemented RAGAS-based LLM evaluation frameworks measuring faithfulness, answer relevancy, context precision, and recall — built Prometheus and Grafana monitoring dashboards tracking response quality drift, API latency, and token utilization across all production GenAI services.
- **REST APIs & microservices for AI:** Built FastAPI Python microservices and REST API layers connecting LangGraph agents, LangChain RAG pipelines, and MCP services with enterprise applications — standardized via OpenAPI/Swagger documentation with OAuth 2.0 security and IAM-governed AWS access controls.
- **AWS cloud-native AI deployment:** Deployed GenAI services on AWS Lambda, SageMaker, ECS, and Bedrock — built serverless document ingestion pipelines on S3/SQS, DynamoDB vector-adjacent feature stores, and CloudWatch observability for production AI reliability at financial services scale.

- **AI workflow automation & CI/CD:** Containerized all GenAI services with Docker and Kubernetes — automated AI model evaluation, testing, and production deployment through Azure DevOps CI/CD with quality gates, enabling zero-downtime releases and full AI lifecycle automation.
- **Polyglot services & frontend:** Built backend services in Python, Java, and Golang with Angular and React.js frontends, exposing REST and GraphQL APIs over PostgreSQL, SQL Server, and DB2 and integrating GenAI capabilities into enterprise applications.

AI Engineer / ML Engineer | Veterans Affairs Department | Jan 2023 – Nov 2025 | Washington, D.C.

- **LangChain RAG & LLM applications:** Developed Python LangChain RAG applications with OpenAI APIs for VA enterprise knowledge management — built document retrieval, semantic search, and LLM-powered response pipelines serving veteran services workflows, reducing manual document lookup time across federal systems.
- **Multi-agent AI systems on AWS:** Built FastAPI Python microservices deploying AI/ML and LLM workloads across AWS SageMaker, Lambda, ECS, and EKS — implemented agent orchestration patterns with IAM least-privilege and OAuth 2.0 security meeting federal FISMA compliance requirements.
- **NLP & document intelligence pipelines:** Developed NLP text classification, named-entity recognition, and semantic document processing pipelines for VA veteran records — using Hugging Face Transformers for model training and SageMaker for scalable batch inference on large federal document corpora.
- **AI model monitoring & drift detection:** Built CloudWatch and Prometheus monitoring dashboards tracking LLM response quality, model accuracy, prediction distribution drift, and API latency — integrated automated alerting that triggered retraining jobs when production AI model performance degraded below defined SLA thresholds.
- **Production AI deployment & MLOps:** Containerized all Python and AI/ML applications with Docker — automated testing and production deployments through Jenkins and GitHub Actions CI/CD with MLflow experiment tracking and model versioning for reproducible, governed AI lifecycle management.
- **Distributed data processing for AI:** Built PySpark, AWS Glue, and Airflow ETL pipelines consuming federal SQL data sources — automating data transformation and loading workflows supporting LLM fine-tuning data preparation and AI/ML model training at scale.

Data Scientist / AI Engineer | Radixweb | Jun 2020 – Dec 2021 | India

- **LLM & AI model integration:** Built FastAPI REST APIs exposing ML and early LLM models for real-time prediction and enterprise AI integration — containerized with Docker and deployed on AWS SageMaker; tracked experiments with MLflow for reproducible AI model development and versioning.
- **NLP & text modeling:** Built NLP text classification, semantic similarity, and embedding-based retrieval pipelines using Hugging Face Transformers and Scikit-learn — developed foundational retrieval-augmented response systems for enterprise knowledge applications preceding modern RAG frameworks.
- **Prompt engineering & model evaluation:** Designed prompt templates and evaluation frameworks comparing model variants — measured statistical significance (precision, recall, F1, ROC-AUC) and applied hyperparameter optimization and cross-validation to improve AI model generalization and production accuracy.
- **Python AI development & data engineering:** Built reusable Pandas, NumPy, SQL, and PySpark pipelines for data preprocessing and feature engineering — automated workflow scheduling with Apache Airflow and implemented CI/CD pipelines with Jenkins for automated AI model validation and release management.

Data Science Intern | Radixweb | Jun 2019 – Jun 2020 | India

- Built Python data preprocessing pipelines and foundational ML models (Scikit-learn); developed NLP text processing workflows; performed EDA and feature engineering; containerized scripts with Docker; used Git for version control; participated in Agile sprint reviews and model evaluation discussions.

EDUCATION & CERTIFICATIONS

M.S. Computer Science — Texas A&M University-Kingsville | Jan 2022 – Dec 2022 | GPA: 4.0/4.0

B.Tech. Computer Science & Engineering — Vignan's University, India | Jul 2016 – May 2020

16 Anthropic Certifications (Claude API, MCP, Agentic AI, RAG, Prompt Engineering, Tool Use, Multi-Agent Systems, Production AI Deployment) • AWS Cloud Practitioner • TensorFlow Developer Certificate